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Electronics L...

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Grounded in 12 sources

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AN

Adam Nowak

adam.nowak@...

...

temperature efficiency

esc

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11 results · 0.04 s

DOCUMENTS 2

Laboratory Report · 2. Theoretical Background

Photovoltaic **efficiency** decreases as **temperature** increases. The open-circuit voltage falls by roughly 2.2 mV per kelvin...

Draft Conclusions · paragraph 2

The **temperature** coefficient of **efficiency** agrees with the literature within 0.05 %/K.

SOURCES 3

smith_2025.pdf · page 14

...conversion **efficiency** fell by 0.41 % per kelvin over the 25–75 °C range **temperature**...

lecture_05.pdf · page 22

...**efficiency** decreases roughly linearly with cell **temperature**, ≈ -0.4 %/K for silicon...

datasheet_cell.pdf · page 2

Temperature coefficient of **efficiency**: -0.38 %/K (**temperature** range 25–65 °C)

AI CONVERSATIONS 2

What is the evidence that efficiency falls with temperature? · Kasia · 2 supporting, 1 contrasting

Strong **efficiency** evidence in 2 sources; one informal note disagrees at 30 °C **temperature**...

ANALYSES 2

Analysis #16 · Voc vs temperature, linear fit · sample_data.csv

columns: **temperature**, voltage, current, **efficiency**

PREVIEW

smith_2025.pdf · page 14

3.2 Temperature dependence

Between 25 °C and 75 °C the mean efficiency decreased monotonically for every cell.

Across the five cells, **conversion efficiency fell by 0.41 % per kelvin over the 25–75 °C range** ($\sigma = 0.03$ % K⁻¹).

...

Open page

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Electronics Lab — Team 4 > Overview > Settings

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KW

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Settings

General

Members

Archived items

General

Everyone in the workspace sees this name and description.

Workspace name

Electronics Lab — Team 4

Description

Save changes

Archive workspace

Removes the workspace from everyone's active list. You can restore it from Archived workspaces any time.

Archive workspace...

Editor sees

Name and description as read-only text. No archive section, no member controls.

Viewer sees

No Settings entry at all. Settings are reached from the workspace menu, which is hidden.

Archive Electronics Lab — Team 4?

Archiving removes this workspace from the active workspace list.

No files or documents will be deleted.

Sources, analyses, the Laboratory Report and every version stay exactly as they are.

Cancel

Archive workspace

E

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Settings

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Adam Nowak

adam.nowak@...

...

Your settings

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Profile
- ☀️

Appearance
- 🔑

Account

Profile

AN

Avatar colour

●

●

●

●

●

Upload photo

Name

Email

Adam Nowak

adam.nowak@uni.edu

Save profile

Appearance

Dark uses warm graphite and muted versions of the same five colors.

☀️ Light

✓

🌙 Dark

🖥️ System

Account

Signed in as adam.nowak@uni.edu on this device.

🔑 Log out

QUESTION

What factors most strongly influenced the measurement error?

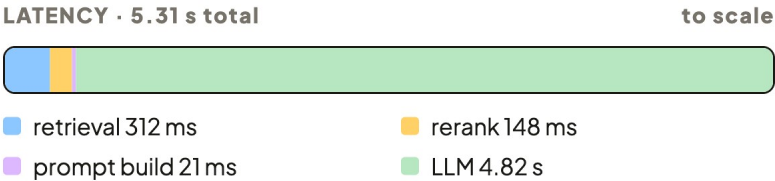
Electronics Lab — Team 4

Michał K.

scope: all 12 sources

today 11:48:02

Model	primary-llm-2026-08
Prompt template	ask_workspace · v3.2.1
Temperature	0.2 · top-p 0.9
Retrieval latency	312 ms
Rerank latency	148 ms
LLM latency	4.82 s (first token 0.9 s)
Context tokens	5,812 in · 612 out
Estimated cost	\$0.0213
Retrieval	hybrid · k = 24 → rerank 12
Cache	miss

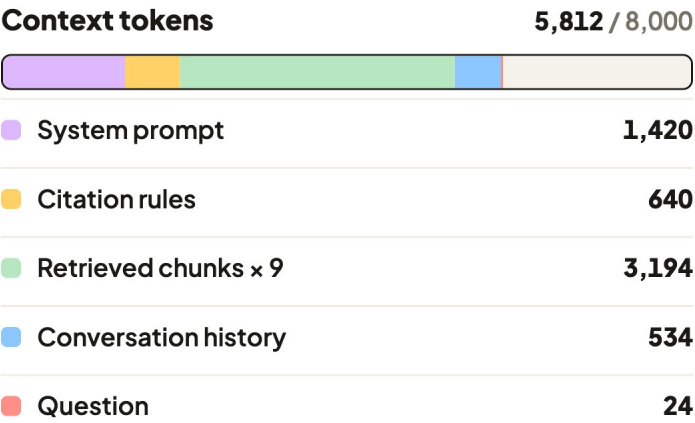


Retrieved chunks · 24 retrieved, 12 reranked, 9 in context

hybrid: vector + keyword

rerank threshold 0.50

#	CHUNK	SOURCE · LOCATION	SNIPPET	RETRIEVAL	RERANK	NEW ORDER	TOKENS	CONTEXT	
			Current drops from 2.40						
			The multimeter resolution is 0.01 Ω						
			Contact resistance of 0.2–0.4 Ω is						
		lecture_05.pdf	Cell temperature rose about 4 °C during the		0.742	0.803	4	402	in
		smith_2025.pdf	Error probably from the						
			Plate and cell temperatures agreed to within 1 K at		0.718	0.771	5 ▲1	377	in
			Record each reading after the						
			The series resistance dominates the high		0.688	0.611	8 —	331	in
			Temperature coefficient of efficiency		0.671	0.583	9 —	290	in
			Lab 3 outline: objective, set-up, measurements		0.655	0.402	11 ▼1	210	dropped · <0.50
			Heterojunction modules lose ≈ 0.27 %/K over the same		0.649	0.511	10 ▲1	355	dropped · budget
			High-frequency rows 1–13: current between 19.9 and 3.1 mA...		0.640	0.377	12 —	301	dropped · <0.50
		measurements.xlsx	measurement_01 · rows 1–13						



Answer check

- ✓

Largest deviation at low frequencies

ch_0142 · ch_0377
- ✓

Contact resistance of 0.2–0.4 Ω

ch_0261
- ✓

Temperature drift ≈ 4 °C

ch_0518
- ⌚

Cable length disputed — no data

ch_0090

citations 4/4 resolved unsupported 0

Prompt · ask_workspace v3.2.1

Copy

[system] Answer only from the sources below. Cite every claim as [n]. If a claim has no source, write "No evidence found". [sources] 9 chunks · 3,194 tokens [history] 2 turns · 534 tokens [question] What factors most strongly...